

Pavan Kumar Nayakanti

AI Production Infrastructure Leader | ModelOps | GCP + Azure | Run-the-Bank | Financial Services

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SUMMARY

20+ year AI production infrastructure and platform engineering leader with a career built inside the world's largest financial institutions - ADP, Charles Schwab, BMO, RBC, and Microsoft. Deep expertise in Run-the-Bank (RTB) and Change-the-Bank (CTB) delivery, ModelOps governance, multi-cloud AI infrastructure (GCP, Azure, AWS), and high-availability production operations for regulated environments. Deployed GenAI production systems on AWS Bedrock and Azure; scaled SRE teams from 4 to 40 engineers; enforced SOX/GDPR-compliant change management across 250+ applications. GCP Professional Cloud Architect, Azure Solutions Architect Expert, and AI Engineer Associate certified.

KEY ACHIEVEMENTS

 AI Production Platform (BMO) Deployed GenAI production platform on AWS Bedrock + Azure: LLM-powered risk summarisation, Q&A inference, and governed BCRG/LTS pipelines for a top-5 North American bank.	 RTB/CTB Delivery at Scale Designed and enforced Run-the-Bank / Change-the-Bank pipeline segregation across 250+ apps at RBC Insurance, achieving SOX-compliant continuous delivery with zero prod incidents.	 SRE Team & MTTR Scaled ADP's SRE organization from 4 to 40 engineers across 3 countries; reduced MTTR 40%+ through AIOps-driven automated remediation and blameless postmortem culture.
 Deployment Automation Built modeas: 200+ deployments/hour, 10,000+ manual ops hours eliminated annually, 600% deployment cadence improvement (ADP). App onboarding reduced from 5 weeks to 5 hours (RBC).		

EXPERIENCE

Sr. Specialist - Cloud Practice & SRE Lead

IRIS Software - India

Sep 2020 - Present | Clients: ADP, Charles Schwab, BMO

- Deployed **GenAI production platform on AWS Bedrock and Azure for BMO**: LLM-powered risk summarisation, interactive Q&A, and governed BCRG/LTS inference pipelines - consolidating fragmented AI initiatives into shared, monitored production infrastructure.
- Implemented **ModelOps governance practices** for BMO's AI workloads: model lifecycle management, performance monitoring, autoscaling of inference services, and security guardrails for model APIs and training datasets.
- Operationalized **multi-cloud AI services on Azure and AWS** - scaling AI usage securely across the enterprise with IAM controls, least-privilege access, and encryption at rest and in transit.
- Defined and governed **SLOs, SLIs, and error budgets** across three global banking clients (ADP, Charles Schwab, BMO); delivered SLA-backed platform reliability to VP and C-suite stakeholders.
- Deployed **Prometheus, Grafana, and Dynatrace** for full-stack observability and AI service monitoring: real-time alerting, distributed tracing, root cause analysis, and automated incident response.
- Reduced **MTTR by 40%+** through AIOps-driven automated remediation (Ansible + Python) and instituted blameless postmortem culture across a 35-member SRE organization.
- Architected **Cloud Platform Maturity Index (CPMI)** - governance framework tracking security, compliance, automation maturity and value realization across three regulated banking clients.
- Scaled ADP's SRE/Platform organization from **4 to 40 engineers** across India, Canada, and US; built Charles Schwab's 25-member Platform Engineering team from the ground up.
- Led **AWS FinOps program for BMO**: cloud cost governance, rightsizing, tagging enforcement - improving cost visibility and reducing waste across the AWS estate.
- Executed **GCP landing zone preparation and workload migration POCs for Charles Schwab**, evaluating GCP-native services for cloud-native AI and application workloads.

Automation Technology Specialist

IRIS Software - Toronto, Canada

Jun 2018 - Sep 2020 | Client: Royal Bank of Canada Insurance

- Embedded **Run-the-Bank (RTB) and Change-the-Bank (CTB)** delivery principles across RBC's engineering model - strict prod/non-prod environment segregation with continuous, frictionless deployment via the Robo Deployment Framework and governed CI/CD pipelines.
- Designed and delivered the **Robo Deployment Framework** - self-service CI/CD orchestration for 250+ applications across 60+ teams; reduced application onboarding from 5 weeks to 5 hours (98% reduction).
- Enforced **SOX/GDPR-compliant change management** across 250+ applications - standardised deployment workflows, audit-ready change evidence, and structured RCA across all production changes.
- Enabled **SAST, DAST, and IAST** scanning as core DevSecOps controls - SonarQube, JFrog Artifactory, Jenkins, and GitHub integrated for shift-left security enforcement.
- Inducted **DORA framework** across RBC Wealth Management and Insurance - Deployment Frequency, Lead Time for Changes, Change Failure Rate, and MTTR as primary engineering health KPIs.
- Implemented **Splunk-based log aggregation** integrated with CI/CD workflows - proactive anomaly detection and rapid production incident response.

Automation Architect

IRIS Software - Toronto, Canada

Feb 2017 - Jun 2018 | Clients: Insurance & Capital Markets

- Designed fault-tolerant, **container-orchestrated deployment pipelines using Kubernetes and Helm** for zero-downtime rollouts across capital markets platforms.
- Pioneered **Cognitive and Intelligent Process Automation (CPA/IPA)** for regulated financial services - automating 80+ applications and 250+ components across heterogeneous stacks.

Technology Manager

TechMahindra Pvt. Ltd. - Noida, India

Jul 2015 - Feb 2017 | Client: Adobe

- Led cross-functional global engineering teams on large-scale enterprise delivery; architected zero-downtime business application migration during Adobe's data centre relocation.

Sr. SDET - QA, Automation & Production Engineering

Microsoft Corporation - India

Jul 2007 - Dec 2014 | Sales & Marketing Platforms (US, EMEA, APAC)

- Held dual mandate as Sr. SDET and **Production Engineering owner** for global Sales & Marketing platforms - sustained **99.99% uptime for 7 years** across US, EMEA, and APAC; owned all non-functional objectives (availability, performance, scalability, security).
- Built QA automation frameworks integrated with CI/CD; managed production incident response, RCA, and infrastructure administration across Windows and Linux; collaborated with product teams in Agile to define reliability targets and error budgets.

Senior Software Testing Engineer

Semantic Space Technologies - India

Jan 2005 - Jun 2007

- L3 Analyst for Statusmart application supporting 3,000+ sales team members at CFC (Bank of America); owned release management, QA, and ETL development.

Software Consultant

Defense Electronics Research Laboratory (DRDO) - India

Oct 2002 - Dec 2004

- Software consulting for defence-grade applications requiring high reliability and security standards.

EDUCATION

Masters in Artificial Intelligence Woolf University	2024
General Management Program (GMP) ISB Hyderabad	2022
M.Sc in Computer Science Osmania University	2002

CERTIFICATIONS

Google Cloud Certified - Professional Cloud Architect (PCA)

Google Cloud - 2026

Microsoft Certified: Azure Solutions Architect Expert (AZ-305)

Microsoft

Microsoft Certified: Azure AI Engineer Associate (AI-102)

Microsoft

Microsoft Certified: Azure Data Scientist Associate (DP-100)

Microsoft

AWS Certified Solutions Architect - Associate (SAA-C01)

Amazon Web Services - 2026

Certified SRE (iSRE)

DevOps Institute